Efficient Mobile Deployment of Large Language Models with Quantization, KV Cache Optimization, and Interactive UI

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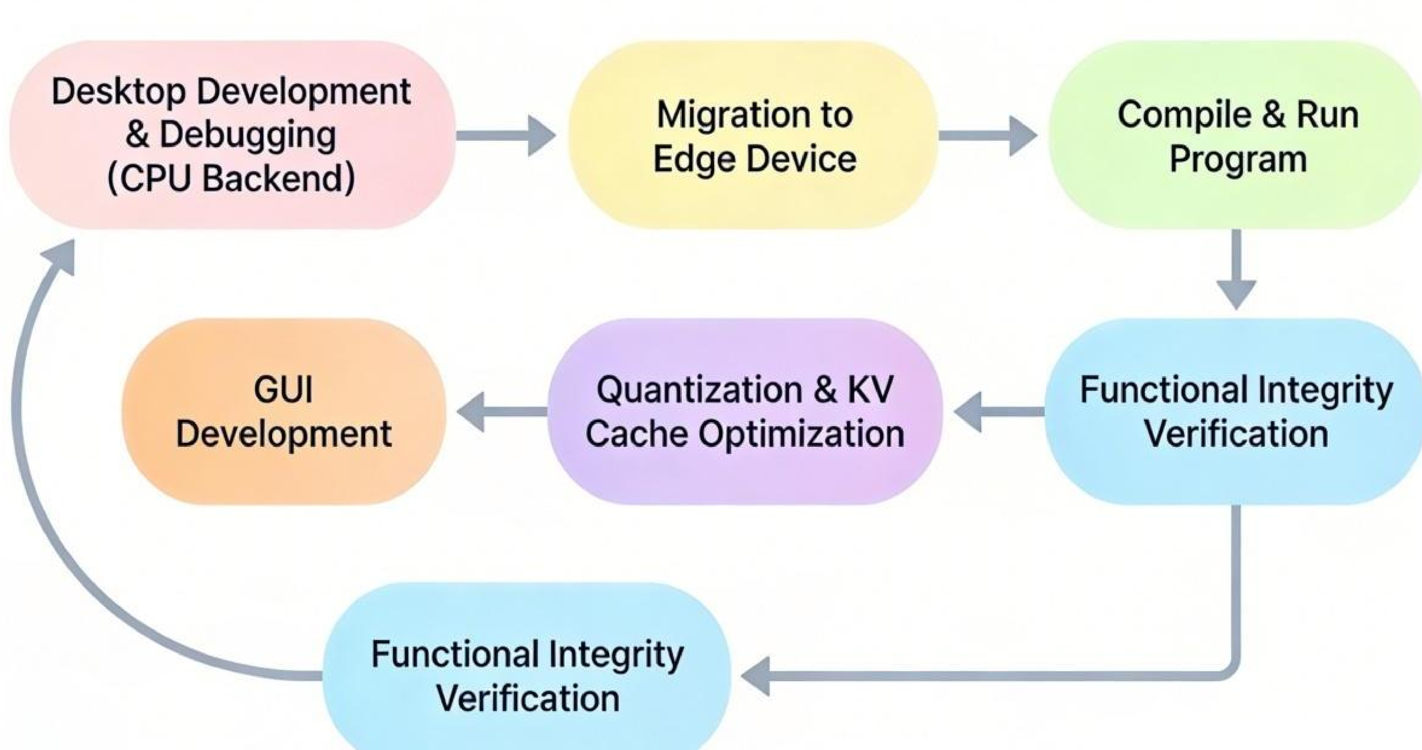
1. Abstract

Large Language Models (LLMs) have achieved impressive performance in a wide range of natural language tasks. However, most existing LLM applications rely on **cloud-based deployment**, which introduces challenges such as high latency, increased cost, and potential privacy risks. Deploying LLMs directly on mobile devices is an attractive alternative, but it is constrained by **limited computational resources and memory capacity**. This project investigates practical techniques to enable efficient and interactive LLM inference on **mobile hardware**.

2. Contribution

- ➢**Deployment of quantized large language models** on mobile devices using llama.cpp
- ➢Application of **KV cache optimization techniques**, including SnapKV and H2O, to reduce memory usage and improve inference efficiency
- ➢Design and implementation of **an interactive Gradio-based graphical interface** with real-time streaming responses



3. Methodology

System Overview

The proposed system enables on-device inference of large language models through **a lightweight and modular architecture**. A **web-based graphical interface** allows users to interact with the model, while inference is handled locally via an HTTP server powered by llama.cpp. Model optimization techniques are applied to **reduce memory usage and improve inference efficiency** on mobile hardware.

KV Cache Optimization

During autoregressive decoding, KV cache size grows linearly with sequence length, posing a major challenge for **long-context inference on mobile devices**. To address this issue, we integrate two KV cache optimization techniques:

SnapKV selectively retains representative tokens, reducing redundant key-value storage.

H2O prioritizes high-importance historical tokens while discarding low-impact ones.

Together, these methods significantly reduce memory usage and enable longer conversations under limited resources.

Interactive Graphical User Interface with Gradio

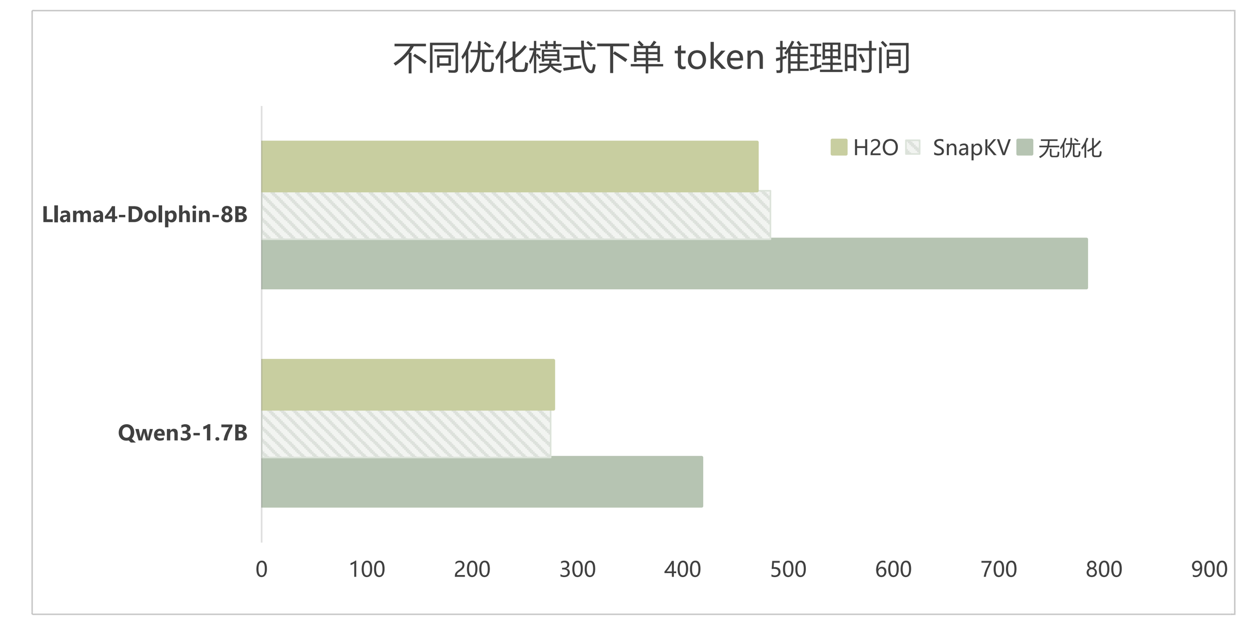
We implement **a lightweight Gradio-based graphical interface** to enable interactive access to on-device large language models on mobile devices.

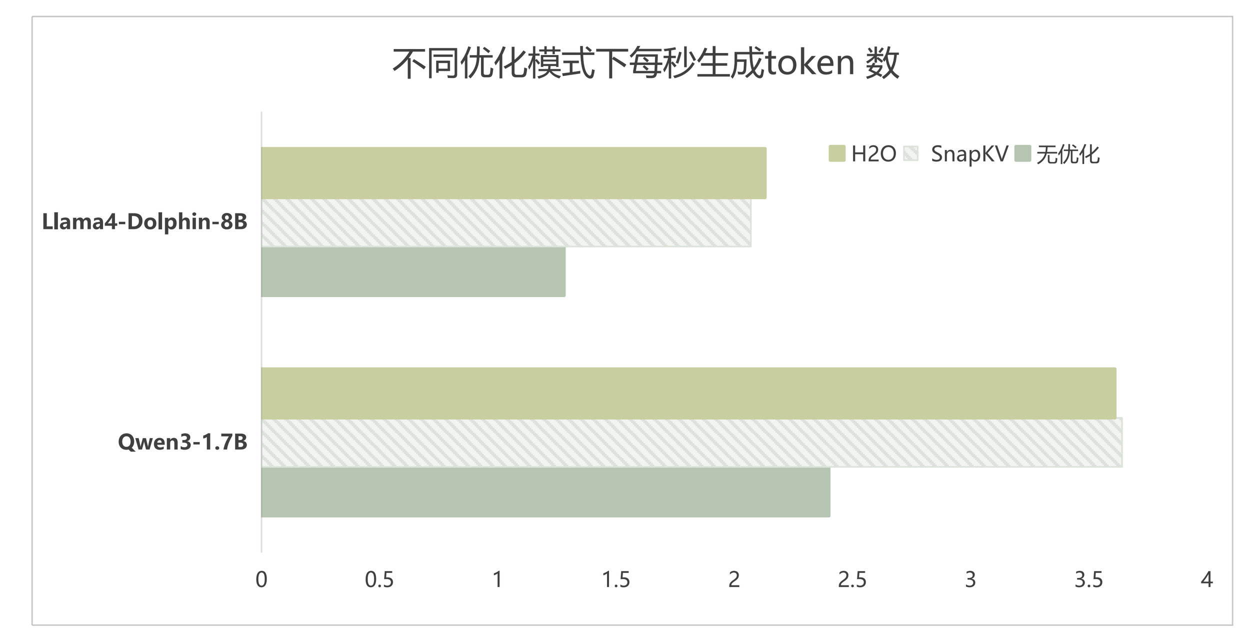
The interface supports **multi-turn chat, streaming responses, and model switching through a mobile browser**.

User inputs are forwarded to a local inference backend (llama.cpp), enabling real-time on-device generation.

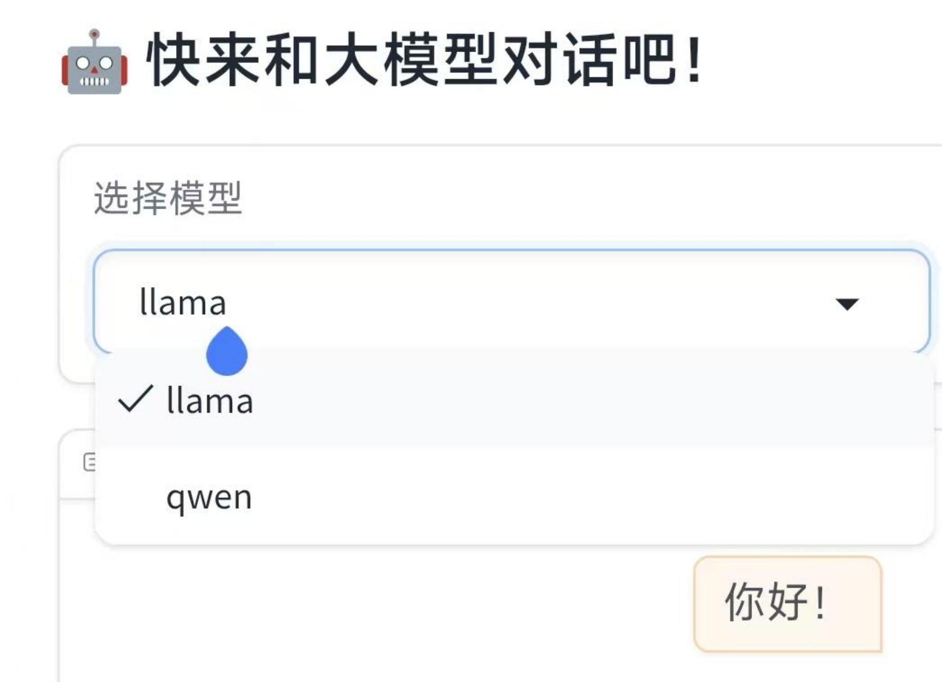
4. Results and Analysis

Experiments on generalization capabilities





Gradio-based Mobile Chat Interface



Gradio-based graphical interface for on-device LLM interaction, supporting streaming responses and model switching on mobile devices.